Project Workstream Allocation: Group J

Task 1: Data Engineering & Feature Synthesis

Primary Focus: Preprocessing, Cleaning, and Consolidation.

- **Cleaning:** Remove duplicate records and handle missing values within the 9,544 pairs.
- **Text Consolidation:** Develop logic to merge unstructured fields (skills, education, experience) into a "Consolidated Text Representation" for both candidates and job postings.
- **Stratification:** Categorize data by the 28 professional domains to ensure the training and test sets are balanced.
- **Deliverable:** A standardized, split dataset (Train/Val/Test) and a preprocessing script for future data.

Task 2: Statistical Baseline & NLP Pipeline

Primary Focus: TF-IDF and Keyword-based Similarity.

- **Vectorization:** Implement a TF-IDF pipeline to transform the consolidated text into sparse vectors.
- **Similarity Logic:** Calculate the Cosine Similarity between candidate and job vectors to establish the baseline "matched_score."
- **Comparative Analysis:** Document instances where the baseline fails due to vocabulary mismatches (e.g., "Deep Learning" vs. "Neural Networks").
- **Deliverable:** A baseline performance report including MSE and Pearson Correlation results.

Task 3: Neural Architecture & Transformer Development

Primary Focus: SBERT Implementation and Regression Tuning.

- **Model Setup:** Initialize the pre-trained all-MiniLM-L6-v2 SBERT architecture.
- **Regression Head:** Add and train a linear layer on top of the transformer output to predict the continuous alignment score (0.0 to 1.0).
- **Fine-tuning:** Optimize hyperparameters (learning rate, batch size) to improve the model's "human-like" understanding of professional context.
- **Deliverable:** A trained Semantic Transformer model and weights.

Task 4: Evaluation Framework & Cross-Validation

Primary Focus: Validation Strategy and Domain Consistency.

- **Metric Suite:** Build an automated script to calculate Mean Squared Error (MSE) and Pearson Correlation (r).
- **Cross-Validation:** Implement the 5-fold CV logic to ensure model stability across different sectors (e.g., Site Engineers vs. Data Scientists).
- **Visualization:** Create high-level data visualizations, such as correlation heatmaps and "Predicted vs. Actual" scatter plots.

- **Deliverable:** A comprehensive evaluation dashboard and final validation report.

Parallelization Strategy

Task	Dependency	Start Time
1. Data Engineering	None	Immediate
2. Baseline Model	Needs Task 1 "Mock" Data	Day 1
3. Advanced Model	Needs Task 1 "Mock" Data	Day 1
4. Evaluation	Needs Task 2/3 Outputs	Day 2